

K-Means Clustering — Short Notes

K-Means is an **unsupervised ML algorithm** that groups similar data points into **K clusters**.

Steps:

1. **Choose K** → number of clusters.
2. **Initialize centroids** → randomly choose K centers.
3. **Assign points** → assign each point to the nearest centroid using Euclidean distance.
4. **Update centroids** → calculate the mean of points in each cluster.
5. **Repeat** → assign → update until centroids stop changing.

Important formulas

Euclidean Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid:

$$\mu = \frac{\sum x_i}{n}$$

WCSS:

$$WCSS = \sum (x_i - \mu)^2$$

Elbow Method

Try different values of **K**, calculate WCSS, and plot **K vs WCSS**.

👉 The **elbow point** is a commonly chosen K because after it, adding more clusters gives only small improvement.

Memory:

Choose K → Assign → Mean → Repeat → Converge

K-Means Clustering — From Scratch

K-Means groups similar data points into **K clusters**.

1. Constructor

class KMeans:

```
def __init__(self, n_clusters, max_iter):
```

```
    self.n_clusters = n_clusters
```

```
    self.max_iter = max_iter
```

- `n_clusters` → number of clusters (**K**)
- `max_iter` → maximum iterations

2. Initialize Centroids

Randomly select K data points as initial centroids.

```
centroids = random.sample(list(X), self.n_clusters)
```

3. Assignment Step

For every point:

- Calculate distance from every centroid.
- Assign it to the **nearest centroid**.

4. Update Centroids

For each cluster, calculate the **mean** of its points:

$$\text{Centroid} = \frac{\sum x_i}{n}$$

Example:

Points = 2, 4, 6

$$\text{Centroid} = \frac{2 + 4 + 6}{3} = 4$$

5. Repeat

Assign points

↓

Calculate new centroids

↓

Check if centroids changed

↓

If changed → repeat

If unchanged → stop

6. Convergence

When centroids stop changing significantly, **K-Means has converged**.

Important

The complete algorithm is:

Choose K → Initialize centroids → Assign → Update → Repeat → Converge

Homework: Elbow Method

Run K-Means with different values of **K**, calculate **WCSS**, and plot:

K vs WCSS

The point where the decrease in WCSS starts becoming much smaller is called the **elbow**, and it is commonly used to choose K .